

Annual Donor Advised Fund Report Study

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Who Are They? What Do They Do?

- Dorothy A. Johnson Center for Philanthropy
 - Helps individuals and organizations understand, strengthen, and advance philanthropy
 - Workshops, Research, Projects
 - Jeff Williams, Director, Consulting Services





What are DAFs?

- Donor advised funds (DAFs) are charitable giving accounts used by donor advisors
- Not all DAF sponsors have their financial data made public by the time the annual report is released
- The Johnson Center has been tasked with predicting the unreleased data and releasing an end of year report





Research Questions

1. How accurate have previous predictions been?
2. How do these predictions hold up across different charity types since 2008?
3. Is there a better way to make predictions for the Donor Advised Fund report?





Our Data and Variables

The Johnson Center is currently responsible for publishing the following data from charitable organizations at the end of each tax year:

- ❖ **Accounts:** quantity of DAFs currently held inside a charitable organization
- ❖ **Contributions:** total amount of money contributed into all DAF accounts under one organization
- ❖ **Grants:** amount of money distributed from DAF accounts
- ❖ **Assets:** balance of all DAF accounts under one organization
- ❖ **Type/Subtype:** classification of charitable organization (e.g. National Charity, University, Hospital, Nonprofit, etc.)

Data Cleaning

We were given two spreadsheets containing recent and historical data from the Johnson Center, which we merged using Python.

After reformatting and combining, we were left with 23329 rows of data, and while cleaning the data we removed:

- Imputed values
- Entries with N/A values
- Organizations with 0 accounts
- Negative monetary values

We eventually created a spreadsheet with 8066 rows (about 35% of the data) and 8 columns



Aggregate Error





Aggregate Error

- We originally summed up the total for a given variable for all charities before computing the error
- This gave very small error values
- However, it's most important to consider the error at the individual charity level, so we switched our error method

DAF Sponsor	Actual	Predicted
Sponsor 1	400	350
Sponsor 2	270	300
Sponsor 3	330	250
Total	1000	900

$$\text{Error} = \frac{|900 - 1000|}{1000} = 0.10$$

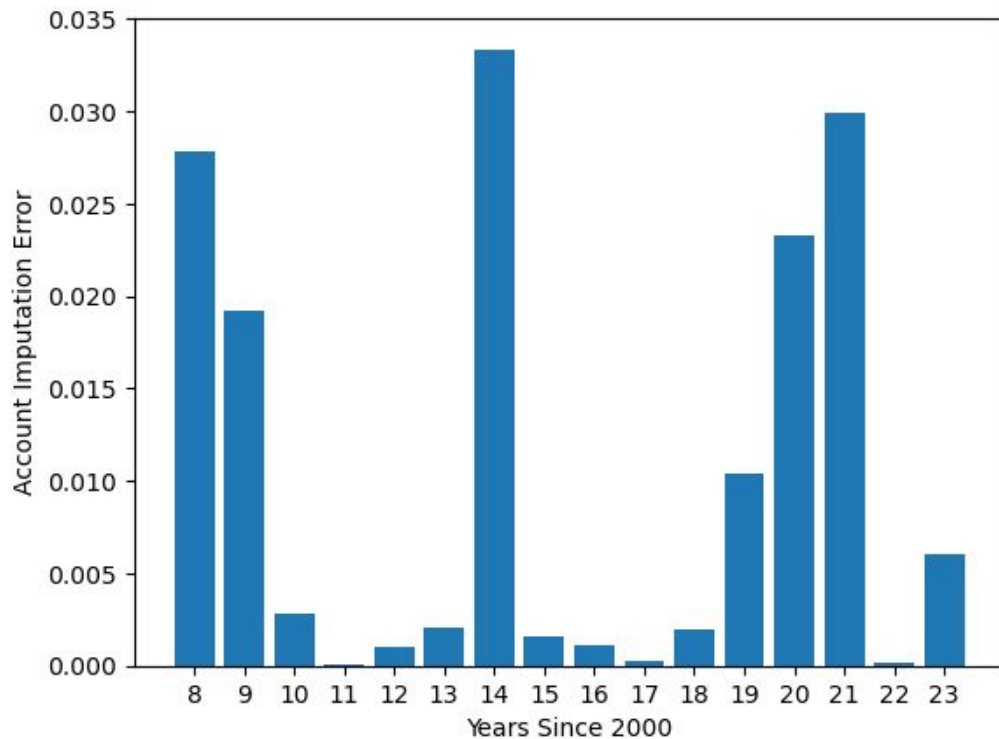


Weighted Average Total Error Over Time

This graph shows many large spikes in imputation error between 2008 and 2023.

Each spike is related to economic recession or economic recovery.

- 2008: Economic Recession
- 2014: Economic Recovery
- 2021: COVID/ Economic Uncertainty



Results and Findings



Johnson Center Imputation Method

of DAF Accounts

- AKA the weighted average growth method
- Group the data by year, type, and subtype
- Calculate the total growth rate for each group
- Apply each group's growth rate to each sponsor in that group

DAF Sponsor	Year 1	Year 2
Sponsor 1	400	500
Sponsor 2	25	25
Sponsor 3	25	25
Sponsor 4	25	25
Sponsor 5	25	25
Total	500	600

$$\text{Group Growth Rate} = \frac{600 - 500}{500} = .20$$

Split Quartile Median Imputation Method

- Group the data by year, type, and subtype, and quartile
- Calculate percent change for each sponsor
- Find the median percent change of quartiles
- Apply to all sponsors in quartile



DAF Sponsor	Percent Change
Sponsor 1	0.21
Sponsor 2	0.25
Sponsor 3	0.29
Sponsor 4	0.32
Sponsor 5	0.34
Sponsor 6	0.38
Sponsor 7	0.41
Sponsor 8	0.43
Sponsor 9	0.56

Results of Methods

To compute error, we took the difference between our predicted values and the actual values and divided by the actual values. We then took the median of all computed errors.

$$\frac{|\text{Actual} - \text{Predicted}|}{\text{Actual}}$$

Weighted Average Growth Method:

Account Error:	7.26%
Contributions Error:	54.53%
Grants Error:	34.16%
Assets Error:	13.90%

Split Quartile Median Method:

Account Error:	1.60%
Contributions Error:	19.59%
Grants Error:	10.90%
Assets Error:	3.17%

Results of Methods

To compute error, we took the difference between our predicted values and the actual values and divided by the actual values. We then took the mean of all computed errors.

$$\frac{|\text{Actual} - \text{Predicted}|}{\text{Actual}}$$

Weighted Average Growth Method:

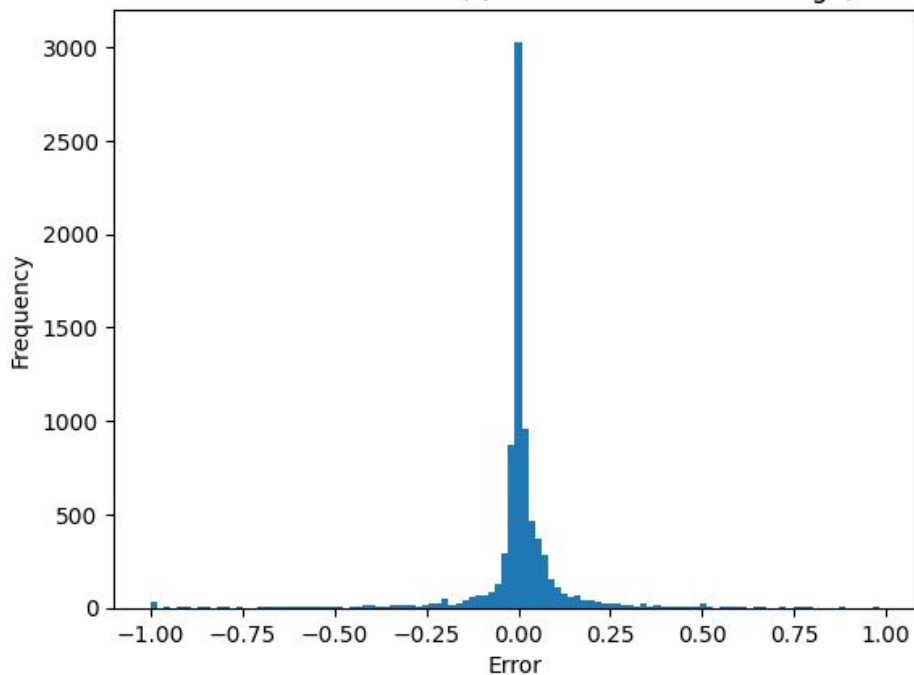
Account Error:	21.70%
Contributions Error:	357.41%
Grants Error:	231.34%
Assets Error:	64.42%

Split Quartile Median Method:

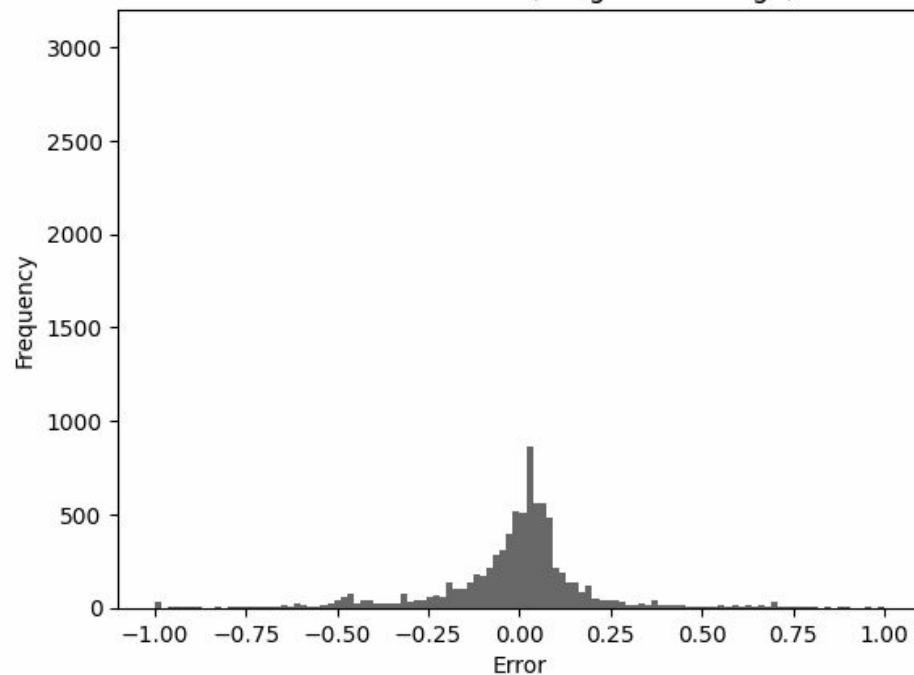
Account Error:	9.15%
Contributions Error:	92.10%
Grants Error:	100.90%
Assets Error:	44.80%

Quartile vs Weighted Average - Accounts

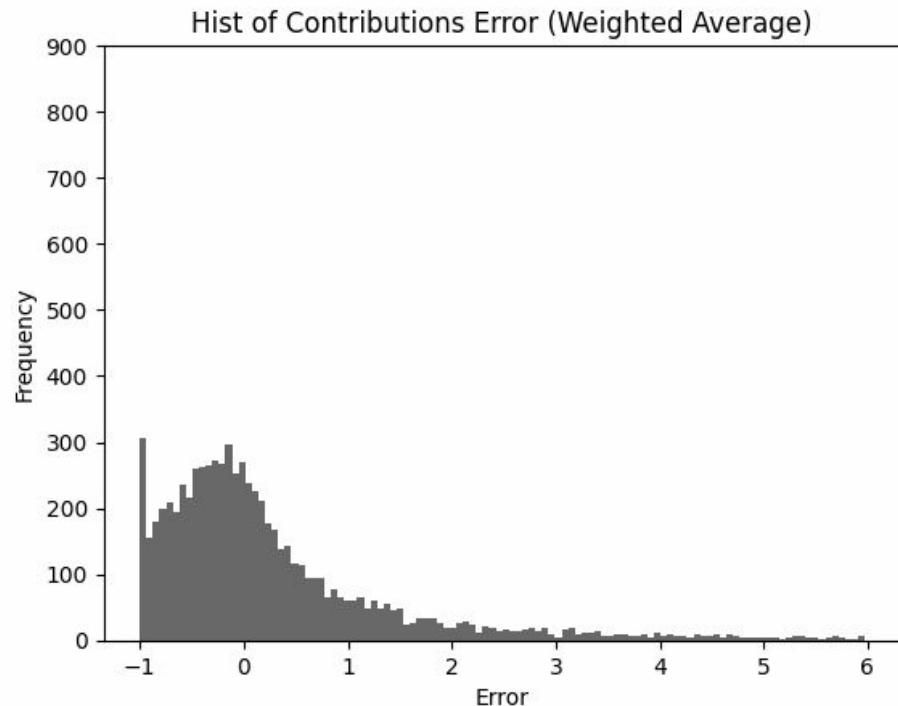
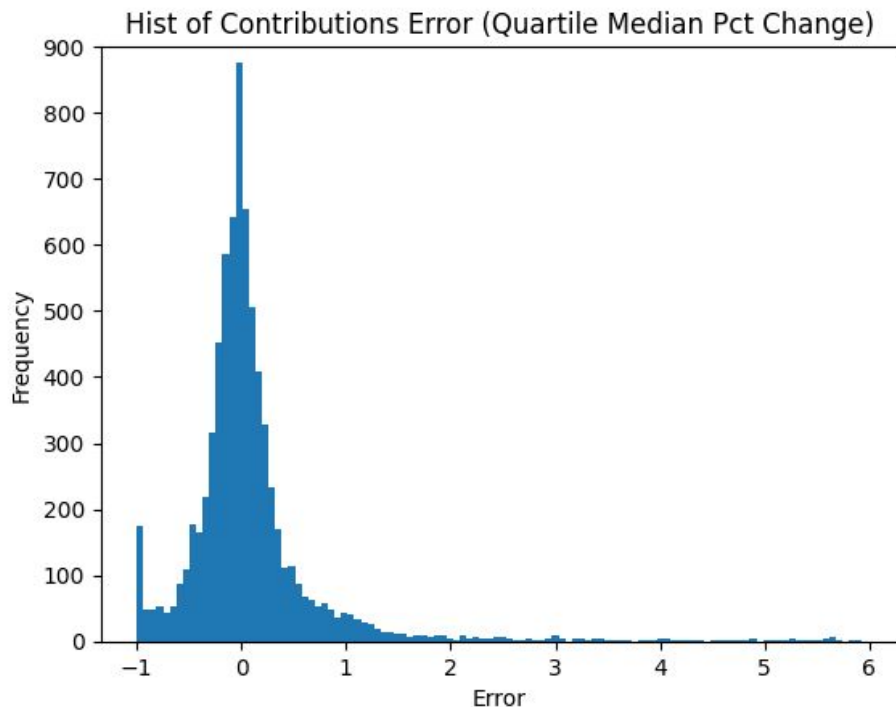
Hist of Accounts Error (Quartile Median Pct Change)



Hist of Accounts Error (Weighted Average)

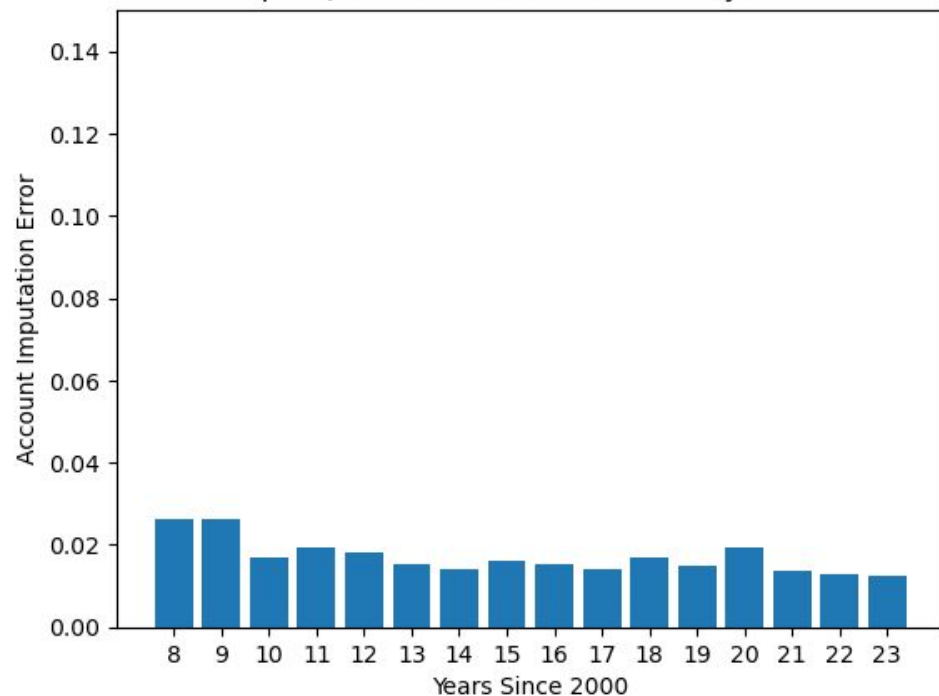


Quartile vs Weighted Average - Contributions

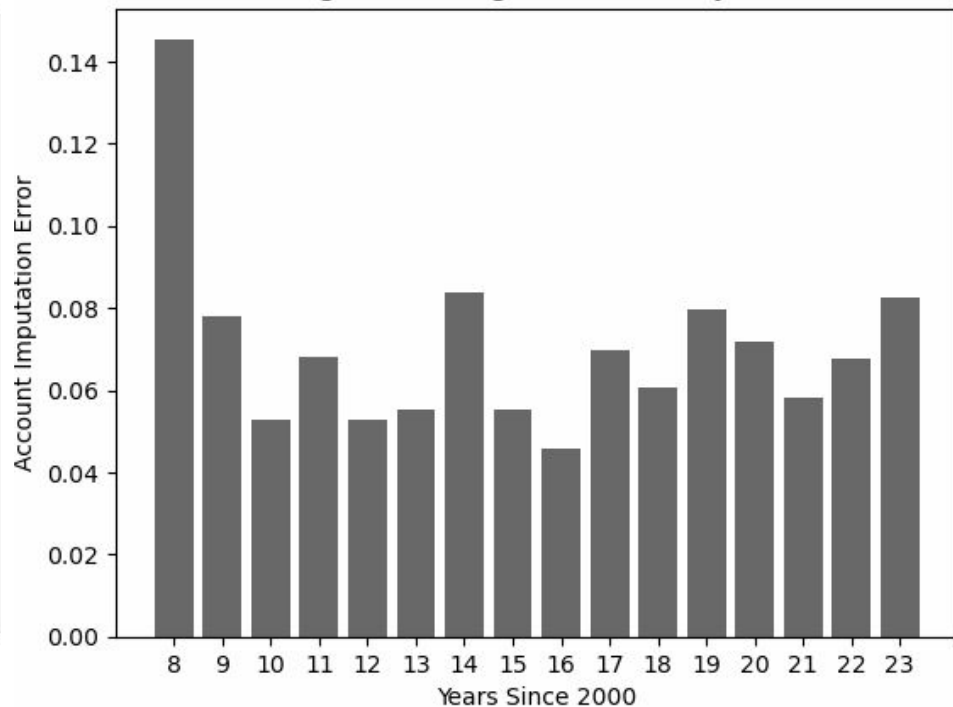


Yearly Error Comparison

Split Quartile Median Method Yearly Error



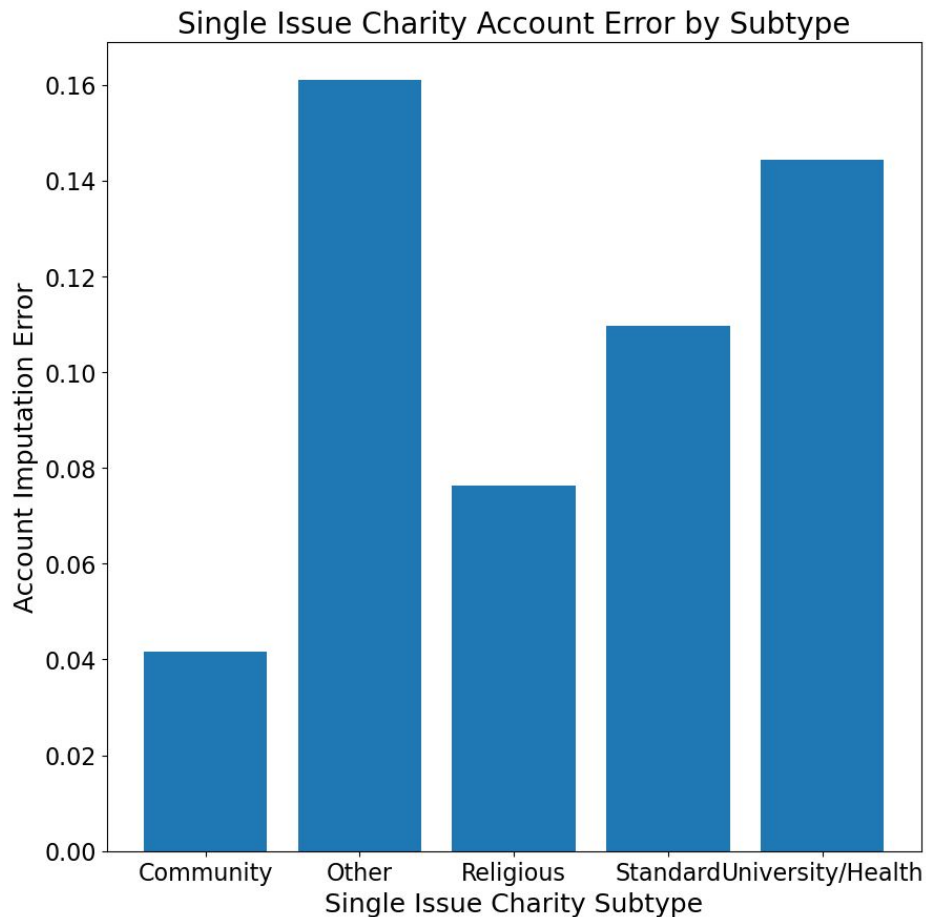
Weighted Average Growth Yearly Error



Weighted Average Error by Type

Various types of DAFs perform better under our imputation methods.

DAFs of type “other” or “healthcare” are infrequent, explaining the higher error in those categories.

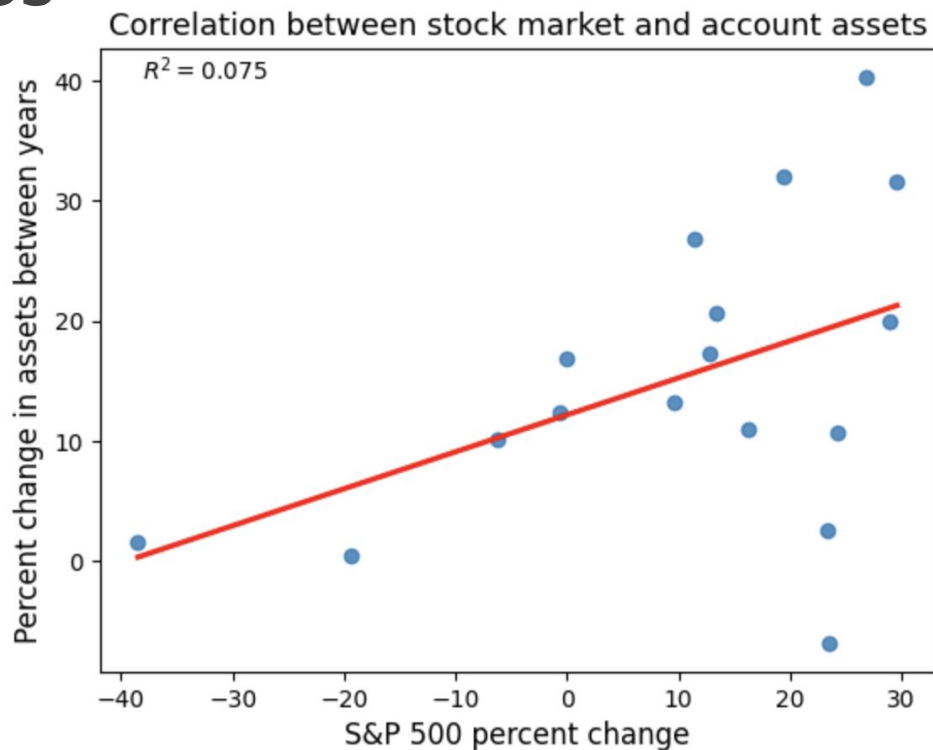


Other Attempted Methods



Suboptimal Imputation Methods

- Linear regression with S&P 500 percent changes as weights
- Winsorizing the data and throwing out the top and bottom 10% of data
- Median percent change



Conclusions





Research Questions Revisited

- ❖ How accurate have previous predictions been?
 - Previous predictions had an average error of 7% for accounts, with much larger errors for other metrics
- ❖ How do these predictions hold up across different charity types since 2008?
 - The prediction errors noticed spikes during years of financial difficulty
 - Less prevalent charities experienced higher errors



Research Questions Revisited

- ❖ Is there a better way to make predictions for the Donor Advised Fund report?
 - By imputing missing data with the split quartile median method, we can vastly reduce the imputation error



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**Thank you for
listening!**

